

TECHBIZ SECURITY

SECURITY OPERATIONS | CYBER DEFENCE | THREAT MONITORING

CYBER SECURITY INTERNSHIP REPORT

WEEK 1 – TASK 2 (ETHICAL HACKING)

Networking Fundamentals for Security Operations

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1. Introduction

This report documents the activities completed under Week 1 – Task 2 of the Cyber Security Internship at TechBiz Security. The task focused on strengthening core networking fundamentals that form the foundation for Security Operations Center (SOC) responsibilities. As part of the task, structured learning modules on TryHackMe covering computer networking basics, the OSI and TCP/IP models, and the Domain Name System (DNS) were completed.

A working knowledge of networking is essential for a SOC analyst because day-to-day security monitoring involves observing network traffic, identifying communicating hosts through IP and MAC addresses, interpreting protocol behaviour, and reviewing DNS activity for signs of unusual or suspicious domain resolution. This task was undertaken as foundational preparation for such SOC activities and does not represent a live monitoring or investigation exercise.

2. Task Objectives

The objectives defined for this task were as follows:

- Understand basic computer networking concepts.
- Understand IPv4 addressing.
- Understand MAC addresses.
- Understand ICMP and the purpose of ping.
- Understand the OSI model.
- Understand the TCP/IP model.
- Understand the relationship between the OSI and TCP/IP models.
- Understand DNS and common DNS record types.
- Perform basic DNS queries using nslookup.
- Connect networking fundamentals with SOC and security monitoring concepts.

3. Networking Fundamentals and SOC Relevance

Networking knowledge underpins nearly every activity performed within a Security Operations Center. Analysts routinely work with IP addresses to identify the source and destination of network traffic, MAC addresses to identify devices on a local network segment, and protocols such as TCP/IP and ICMP to understand how communication takes place between hosts. DNS activity is reviewed to understand which domains a device is attempting to resolve, which is often a key indicator during security monitoring.

Building familiarity with these concepts allows a trainee to recognise normal network behaviour, which is a necessary baseline before unusual or potentially malicious activity can be identified. This task therefore concentrated on establishing that baseline knowledge across IP addressing, MAC addressing, network protocols, ICMP, the TCP/IP suite, and DNS, providing the conceptual foundation required for later, more advanced SOC activities.

4. What is Networking?

The first learning module completed was TryHackMe's "What is Networking?" room, which introduces the fundamentals of computer networking. At its core, networking refers to the ability of devices to communicate and exchange data with one another. This requires each device to be uniquely identifiable on the network, a defined

protocol governing how data is formatted and transmitted, and a physical or wireless medium over which the transmission occurs.

The module covered the basic building blocks of connectivity: network addressing (how a device is identified), protocols (the agreed rules of communication), and data transmission (how information physically moves between devices). These concepts are the starting point for understanding how any subsequent network-level security monitoring or investigation is carried out, since every security event ultimately involves communication between identifiable network entities.

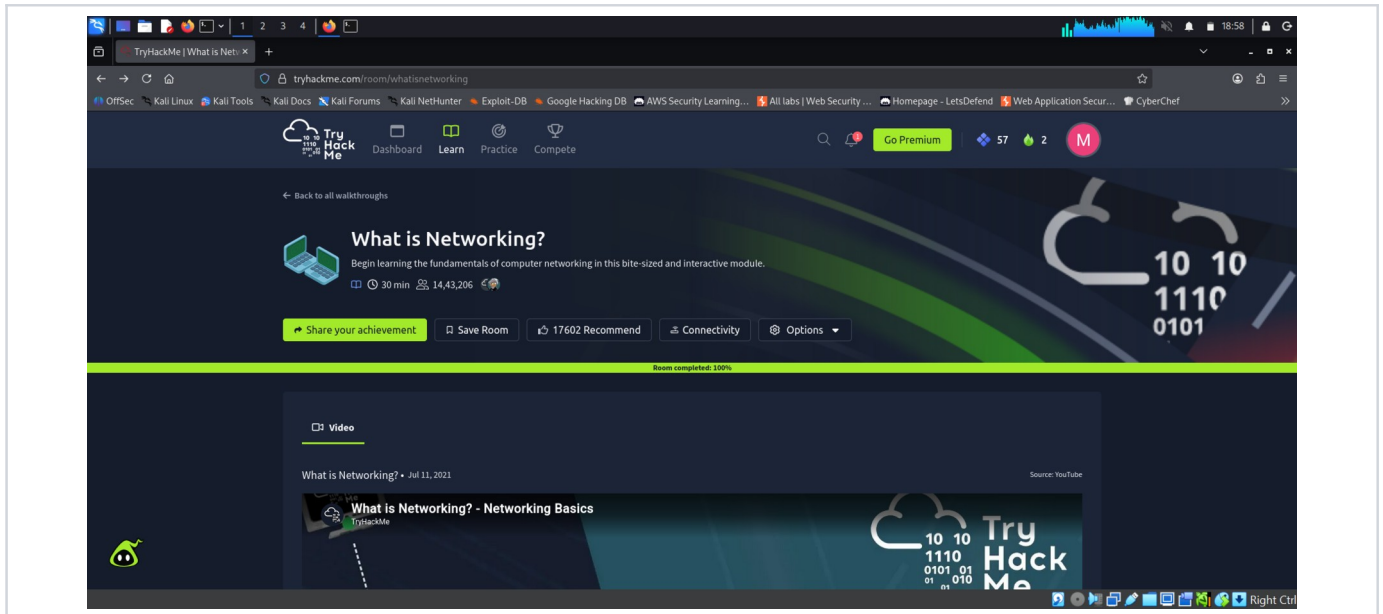


Figure 1: Completion of the TryHackMe “What is Networking?” learning module.

5. Internet Protocol and Network Communication

The Internet Protocol (IP) is the addressing and routing mechanism that allows systems to identify and communicate with other hosts, both within a local network and across the wider internet. Every device participating in network communication is assigned an IP address, which functions similarly to a postal address: it allows data to be correctly routed from a source system to its intended destination.

Within the scope of this task, IP addressing was studied as the mechanism that enables two hosts to locate and exchange data with one another, forming the basis for later topics such as ICMP-based connectivity testing and DNS resolution, both of which rely on an underlying IP address to function.

6. IPv4 Addressing

IPv4 (Internet Protocol version 4) is the addressing scheme most commonly used to identify devices on a network. An IPv4 address is made up of four sections, separated by periods, with each section represented in decimal form. This structure was confirmed during the practical exercise on the “What is Networking?” room, where the number of sections in an IPv4 address was correctly identified as four.

IPv4 addressing is directly relevant to SOC operations for several reasons: it allows an analyst to identify which hosts are communicating on a network, it forms the basis for investigating network events by distinguishing source and destination addresses, and it enables analysis of traffic patterns to determine whether communication is occurring with expected or unexpected hosts. The practical evidence for the IPv4 addressing exercise is presented together with the MAC address activity in Figure 2.

7. MAC Address and Network Identification

MAC stands for Media Access Control. A MAC address is a hardware-level identifier assigned to a network interface and is used to identify devices at the local network level, as opposed to an IP address, which identifies a device at the network (logical) level. MAC addresses are particularly relevant when identifying specific devices on a local network segment, such as during local traffic analysis or when tracing device-level activity.

As part of the practical activity, an interactive lab required spoofing a device's MAC address in order to gain access to a simulated site. Completing this activity correctly returned the flag `THM{YOU_GOT_ON_TRYHACKME}`, confirming that MAC-address-based access control had been demonstrated as instructed by the room. This exercise illustrated how a device's MAC address can act as a form of network-level identification and access control.

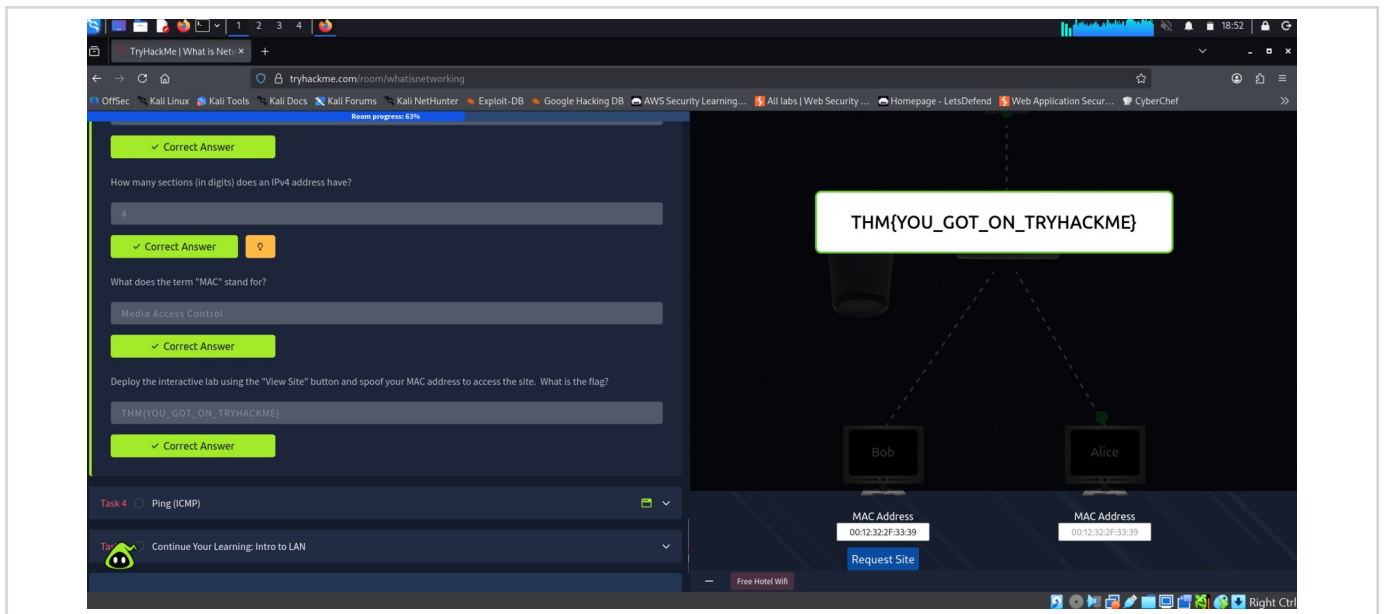


Figure 2: Practical evidence of the IPv4 addressing question and the MAC-address-based access activity.

8. Ping and ICMP

Ping is a network utility used to test the reachability of a host on a network. It operates using the Internet Control Message Protocol (ICMP), which is designed for network diagnostics and error reporting rather than carrying application data. ICMP is a lightweight and commonly encountered protocol, and SOC analysts may observe ICMP traffic during routine network monitoring, connectivity troubleshooting, or while investigating unusual scanning or reconnaissance activity.

During the practical exercise, the command `ping -c 4 8.8.8.8` was executed. This command sends four ICMP echo request packets to the destination address 8.8.8.8 and reports the corresponding echo replies, along with round-trip time statistics. The output confirmed that all four packets were transmitted and received with zero packet loss, and returned the flag `THM{I_PINGED_THE_SERVER}` as evidence that the exercise had been completed successfully.

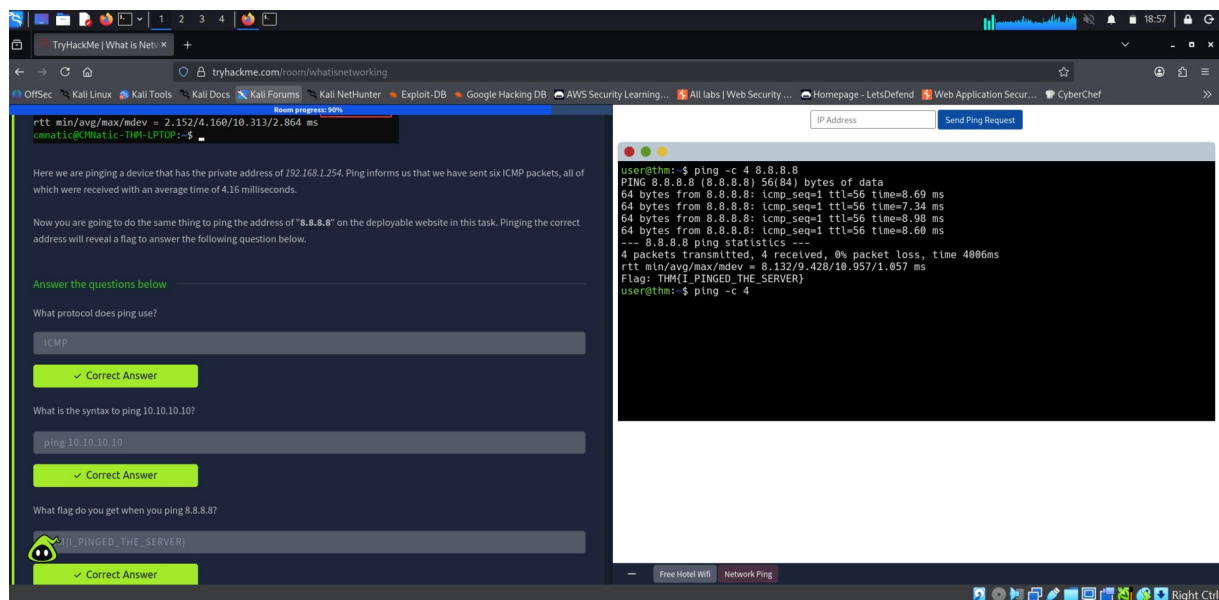


Figure 3: ICMP ping practical demonstrating network connectivity testing using ping -c 4 8.8.8.8.

9. OSI Model

Following completion of the introductory module, the “Introductory Networking” room was completed in full, covering networking theory and basic networking tools, including the OSI model, encapsulation, the TCP/IP model, and the ping utility.

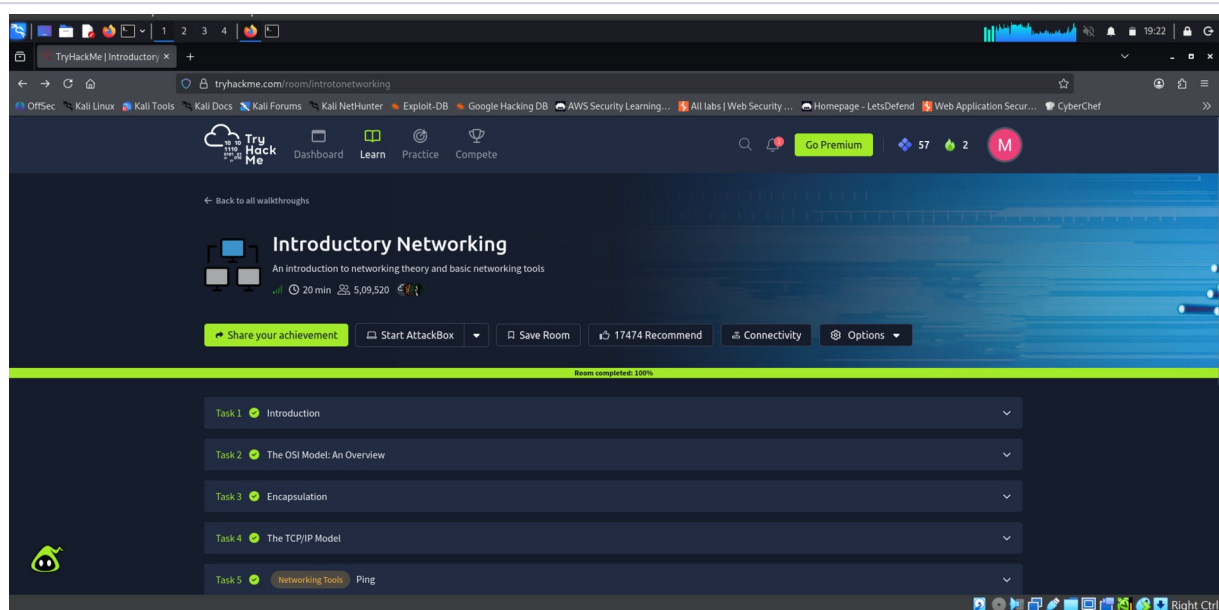


Figure 4: Completion of the TryHackMe “Introductory Networking” learning module.

The Open Systems Interconnection (OSI) model divides network communication into seven distinct layers, each responsible for a specific function. The seven layers, from top to bottom, are as follows:

Layer	Name	Function
Layer 7	Application	Provides network services directly to end-user applications (e.g. HTTP, DNS).

Layer	Name	Function
Layer 6	Presentation	Formats, encrypts, or compresses data into a standardised representation.
Layer 5	Session	Establishes and tracks communication sessions between host and destination.
Layer 4	Transport	Manages end-to-end data delivery (e.g. TCP/UDP communication).
Layer 3	Network	Handles logical addressing and routing (e.g. IP addressing).
Layer 2	Data Link	Handles addressing on the local network segment (e.g. MAC addressing).
Layer 1	Physical	Represents the physical transmission medium and raw bit transmission.

During the practical questions on this room, the layer responsible for verifying that received information has not been corrupted was correctly identified as Layer 2 (Data Link), the layer responsible for formatting data prior to transmission as Layer 2 (Data Link, alongside its data-formatting checks), the layer that transmits and receives raw data as Layer 1 (Physical), the layer that encrypts, compresses, or otherwise standardises data as Layer 6 (Presentation), the layer that tracks communication between host and receiving computers as Layer 5 (Session), and the layer that accepts communication requests from applications as Layer 7 (Application).

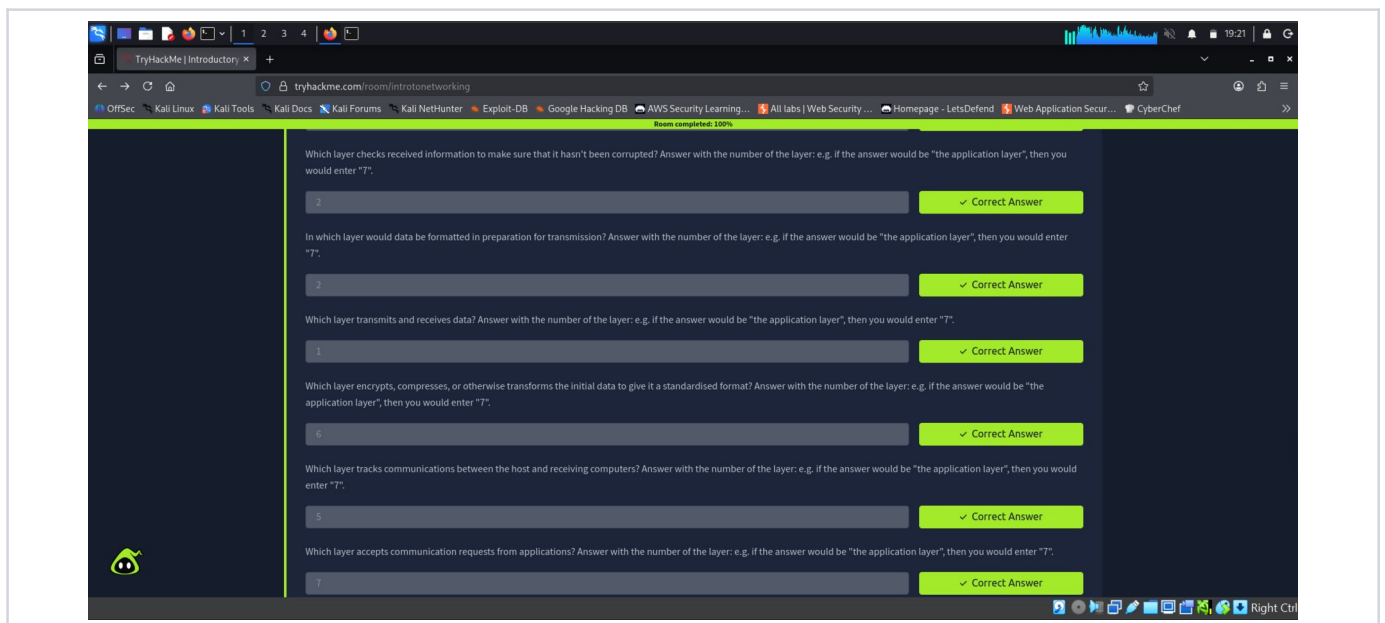


Figure 5: OSI Model learning activity and practical question responses.

From a SOC perspective, the OSI model provides a structured way to reason about where in the communication stack a security event is occurring — for example, application-layer protocols such as HTTP and DNS at Layer 7, TCP/UDP session behaviour at Layer 4, IP-based routing and addressing at Layer 3, and MAC-based local identification at Layer 2.

10. TCP/IP Model

The TCP/IP model is a more condensed, practically-oriented model of network communication, consisting of four layers: Application, Transport, Internet, and Network Interface. Unlike the seven-layer OSI model, the TCP/IP model combines several OSI functions into fewer layers, reflecting how networking protocols are implemented in practice.

The practical questions confirmed the following relationships between the two models: the TCP/IP model was introduced before the OSI model; the Transport layer of the TCP/IP model covers the functionality of the OSI Transport layer; the Application layer of the TCP/IP model covers the functionality of the OSI Session layer; the Network Interface layer of the TCP/IP model covers the functionality of both the OSI Data Link and Physical layers; the Internet layer of the TCP/IP model handles the functionality of the OSI Network layer; and TCP itself was confirmed to be a connection-based protocol.

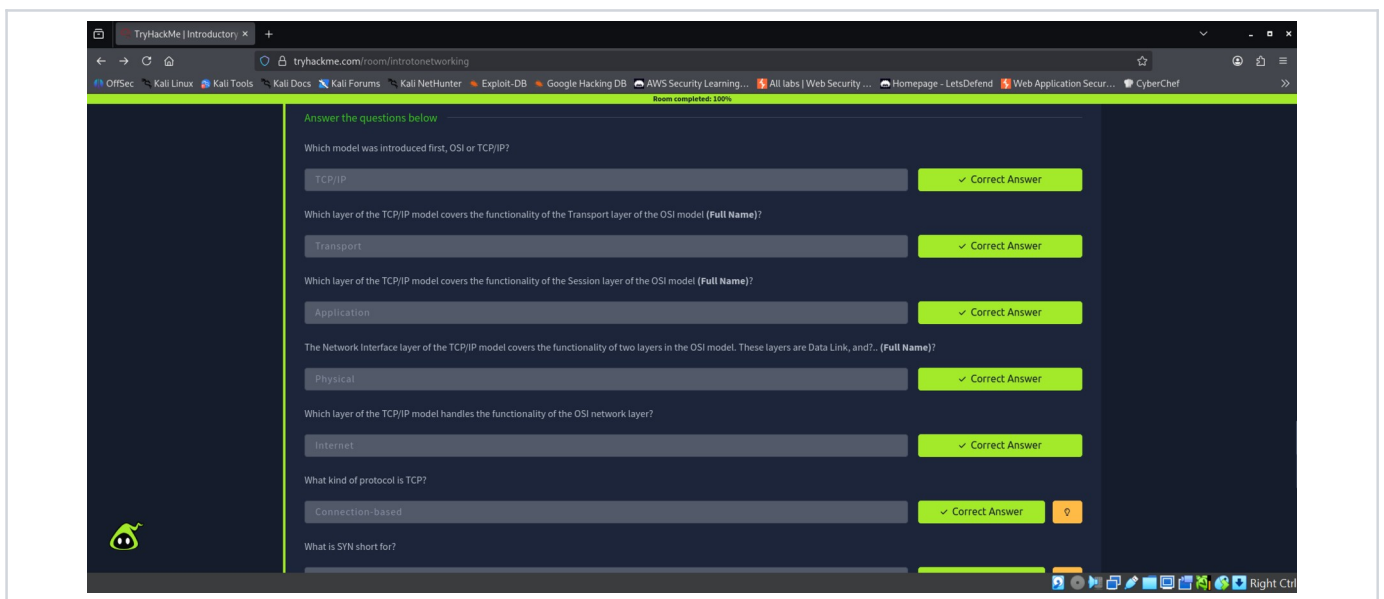


Figure 6: TCP/IP model learning activity and comparison with the OSI model.

11. OSI and TCP/IP Model Relationship

The table below summarises how the seven layers of the OSI model map onto the four layers of the TCP/IP model, based on the relationships confirmed during the practical questions.

OSI Model	TCP/IP Model	Main Function
Application	Application	Network services to applications
Presentation	Application	Data representation and formatting
Session	Application	Session establishment and tracking
Transport	Transport	End-to-end communication
Network	Internet	IP addressing and routing
Data Link	Network Interface	Local network communication
Physical	Network Interface	Physical transmission

As shown above, the TCP/IP model combines several OSI layers into a smaller number of practical layers, most notably merging the Application, Presentation, and Session layers into a single Application layer, and the Data Link and Physical layers into a single Network Interface layer.

12. DNS Fundamentals

The final learning module completed for this task was TryHackMe's "DNS in Detail" room. The Domain Name System (DNS) is the system responsible for translating human-readable domain names into the information required for network communication, most commonly an IP address. Without DNS, users would need to memorise numerical IP addresses in order to access websites and online services.

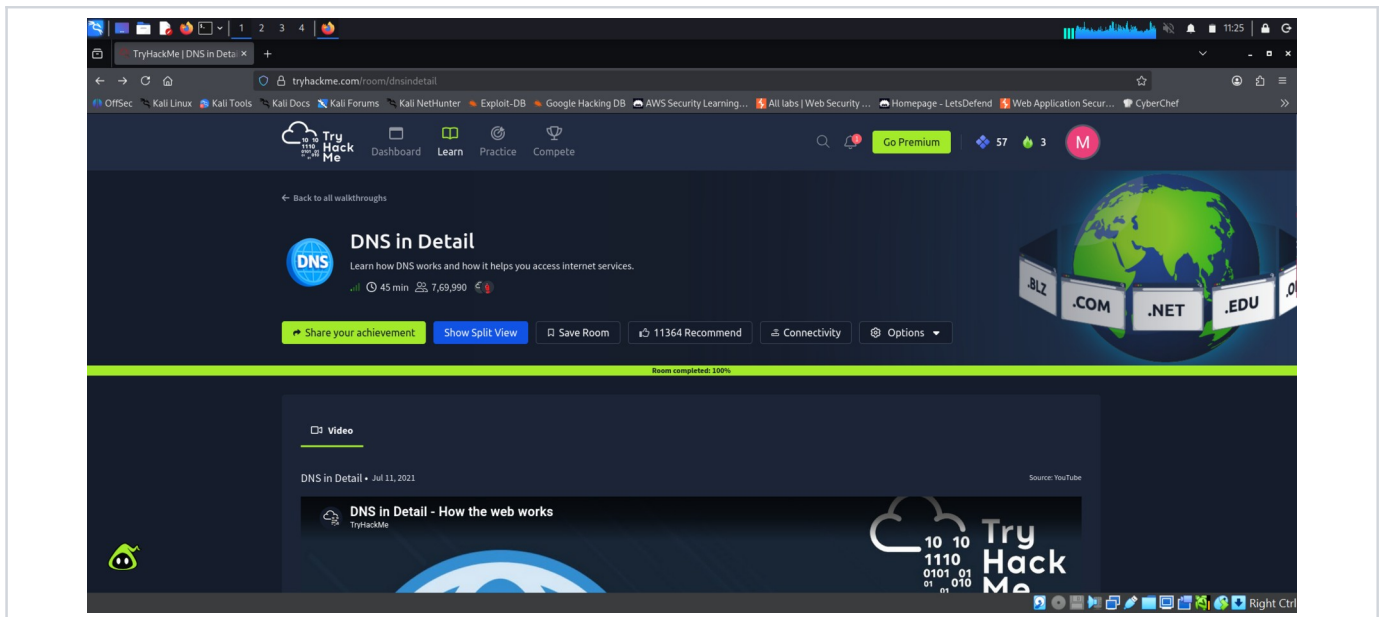


Figure 7: Completion of the TryHackMe "DNS in Detail" learning module.

DNS is highly relevant to SOC operations for several reasons. DNS requests generated by a device can reveal what domains a user or endpoint is attempting to reach, which is a useful indicator of activity during monitoring. Suspicious or unfamiliar domains appearing in DNS traffic may warrant further review, and DNS is frequently involved as a communication channel in malware behaviour. Monitoring DNS resolution activity therefore helps in identifying unusual or unexpected domain lookups. This task did not involve performing malware analysis; DNS was studied purely as a foundational network service.

13. DNS Record Types

The following DNS record types were covered as part of the module and demonstrated during the practical exercise:

Record Type	Description
A Record	Maps a domain or hostname to an IPv4 address.
CNAME Record	Provides an alias, pointing one hostname to another hostname.
TXT Record	Stores arbitrary text information associated with a domain.
MX Record	Specifies the mail server responsible for receiving email for a domain, along with a priority value.

14. Practical DNS Queries

The practical component of the DNS in Detail room made use of nslookup, a command-line utility for querying DNS servers and returning records of a specified type. The following queries were executed:

```
nslookup --type=CNAME shop.website.thm
nslookup --type=TXT website.thm
nslookup --type=MX website.thm
```

The CNAME query against shop.website.thm was used to identify any alias configured for that hostname, and returned a canonical name of shops.myshopify.com. The TXT query against website.thm was used to retrieve text-based record data, and returned the value THM{7012BBA60997F35A9516C2E16D2944FF}. The MX query against website.thm was used to identify the mail server configured for the domain, and returned a numerical priority value of 30 for the associated mail exchanger.

Querying DNS records in this manner is a form of DNS enumeration, and this type of visibility is relevant to security monitoring: understanding which records are published for a domain (such as mail servers, aliases, and text-based metadata) can help an analyst build a picture of an organisation's external infrastructure and identify configuration details that may be relevant during an investigation.

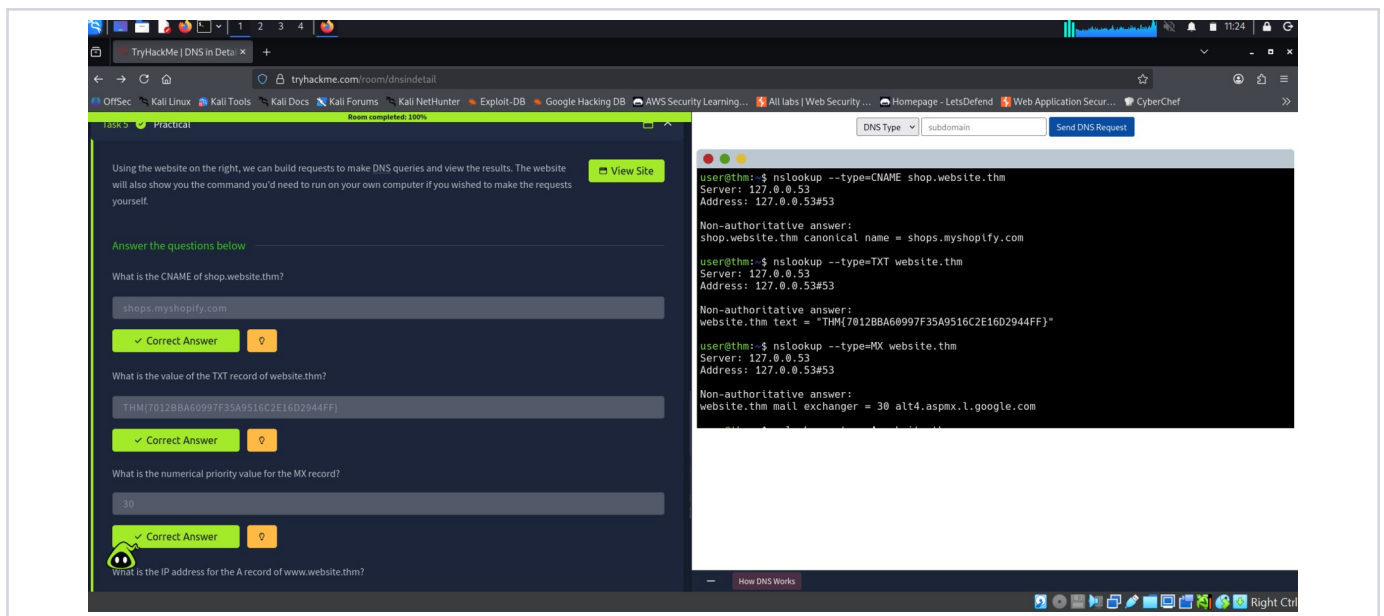


Figure 8: Practical DNS queries using nslookup for CNAME, TXT, and MX records.

15. Practical Evidence and Screenshots

All practical activities described in this report were completed on the TryHackMe platform, and the corresponding screenshots have been presented alongside their relevant sections above rather than repeated separately, in order to keep each screenshot in context with the concept it demonstrates. The index below summarises the full set of evidence for reference.

Figure	Description	Reference
1	Completion of the "What is Networking?" learning module	Section 4
2	IPv4 addressing and MAC-address-based access activity	Section 7
3	ICMP ping practical (ping -c 4 8.8.8.8)	Section 8

Figure	Description	Reference
4	Completion of the “Introductory Networking” learning module	Section 9
5	OSI Model learning activity	Section 9
6	TCP/IP Model learning activity	Section 10
7	Completion of the “DNS in Detail” learning module	Section 12
8	Practical DNS queries using nslookup	Section 14

16. SOC Relevance and Security Applications

The table below links each networking concept covered in this task to its relevance within SOC operations.

Networking Concept	SOC Relevance
IPv4	Identifying source and destination hosts
MAC Address	Identifying devices and interfaces on local networks
ICMP	Monitoring connectivity and unusual ICMP traffic
TCP/IP	Understanding underlying network communication
OSI Model	Analysing network activity by communication layer
DNS	Monitoring domain resolution and suspicious domains
CNAME	Understanding domain aliases and infrastructure
TXT	Identifying domain-associated metadata
MX	Understanding an organisation's email infrastructure

These concepts collectively form the foundational knowledge required for future SOC activities such as alert triage, network traffic analysis, SIEM monitoring, incident investigation, and threat detection. This task did not involve carrying out these advanced activities; it represents the preparatory groundwork on which they depend.

17. Key Learning Outcomes

- Strengthened understanding of core networking fundamentals.
- Understood the structure and purpose of IPv4 addressing.
- Understood the role of MAC addresses in local network identification.
- Learned the purpose of ICMP and the ping utility.
- Learned the seven layers of the OSI model and their functions.
- Understood the structure of the TCP/IP model.
- Understood the relationship and mapping between the OSI and TCP/IP models.
- Learned DNS fundamentals and how domain resolution works.
- Learned the major DNS record types: A, CNAME, TXT, and MX.
- Practised performing DNS queries using nslookup.

- Understood the importance of networking knowledge for SOC operations.

18. Conclusion

Week 1 – Task 2 of the Cyber Security Internship provided foundational networking knowledge required for cybersecurity and SOC operations. Through the completion of structured TryHackMe modules and their associated practical exercises, an understanding was developed of IP addressing, MAC addresses, ICMP, the OSI and TCP/IP models, and DNS, including common DNS record types and the use of nslookup for DNS queries.

Together, these concepts establish an understanding of how systems communicate on a network, which is a necessary foundation for interpreting network-related security events. This task represents preparatory learning rather than applied SOC experience, and provides a basis on which subsequent, more advanced practical tasks in this internship will build.